

## Project Outline

### Mercer's Theorem and Application

1. Mercer Kernels
2. The Operator  $L_x$ 
  - a. Properties of  $L_x$
  - b. Connection to finite dimensional Hilbert Spaces
3. Mercer's Theorem
  - a. A quick sketch of proof
4. Solutions and Applications
  - a. Using symmetry and Boundary Conditions
  - b. The Gaussian Kernel on  $S^1$
  - c. The kernel  $(1 + x^\top y)^d$  on  $S^{n-1}$
  - d. The kernel  $(1 + x^\top y)^2$  on  $\{-1, 1\}^n$
  - e. Kernel tuning by selection of eigenvalue behavior.
5. Derivation of Feature Spaces and RKHS.